

CoRoT-2b

R-2199

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_120905 · created 2026-08-14T00:33:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456175.7467 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.064
Transit depth (Rp/Rs)^2	0.00014 +/- 0.015 [%]
Orbital Inclination (inc)	87.66 +/- 2.07
Airmass coefficient 1 (a1)	0.13 +/- 0.11
Airmass coefficient 2 (a2)	1.63 +/- 0.43
Scatter in the residuals of the lightcurve fit is	79.16 %
Best Comparison Star	#3 - [174, 36]
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.0814 +/- 0.0089

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.064 vs 0.1667 (deviation 1.54σ)
Duration fit vs published	0.0814 d vs 0.095 d (deviation 0.91σ)
Mid-time O–C	-3.5 min
Depth SNR	0.0
Residual scatter	79.16 %

Light curve

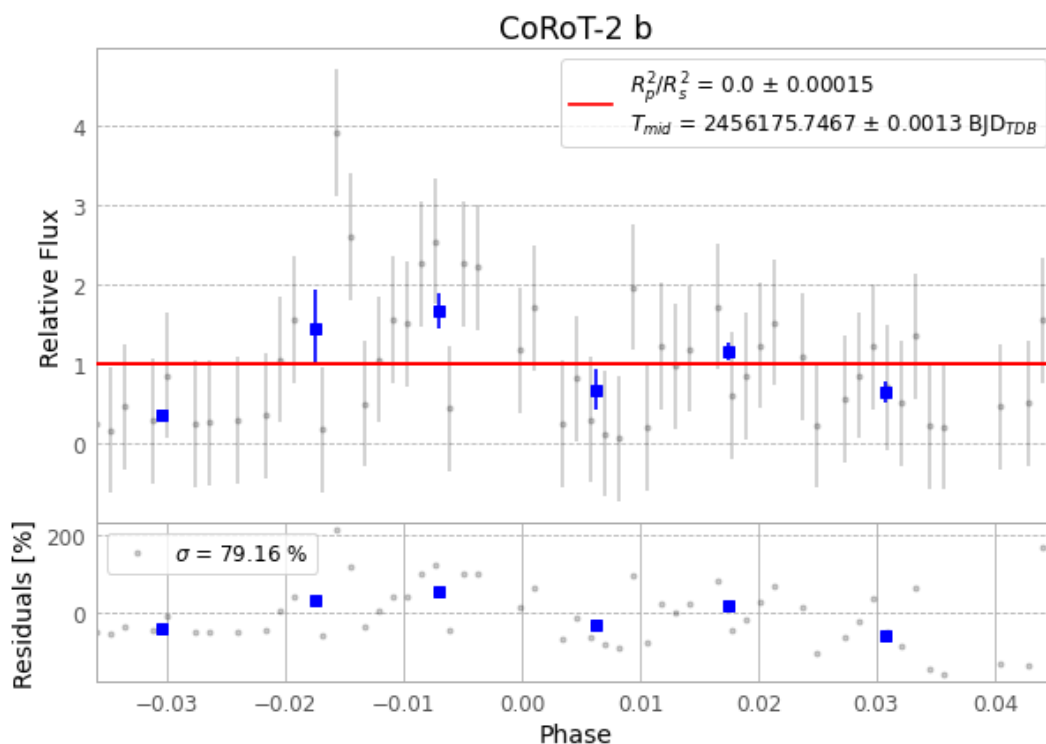


Figure 1 — FinalLightCurve_CoRoT-2 b_2012-09-04.png (SHA-256 24ff668b5607a74c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [443, 261], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 34bfd6979bb14446...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

70 FITS frames (46 MB), CoRoT-2120905041512.FITS → CoRoT-2120905074212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-120905.log (593920b0e8b3...), FinalLightCurve_CoRoT-2 b_2012-09-04.png (24ff668b5607...), FinalLightCurve_CoRoT-2 b_2012-09-04.csv (f47cacdc3069...)

Compute resources

Wall time 169.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 00:33Z.